ECON 5021 Unit 2: SAQ Flashcards

All 17 Canvas SAQs (Lectures 8–12) with the shortest answer that earns the points. 20 min for 5 proofs is ≈ 4 min each. Cover the answer, recite the **hook**, write the answer, then check. Full versions are in `ECON5021_Unit2_Study_Guide.pdf`.

Card 1 — L8 Q1 · mean minimizes MSE

$\mathrm{E}(y) = \mu$. Show the a that minimizes $\mathrm{E}[(y-a)^2]$ is μ.

Hook: add and subtract μ; the cross term dies.

Let $u = y - \mu$, so $\mathrm{E}u = 0$. $\mathrm{E}[(y-a)^2] = \mathrm{E}[(u+\mu-a)^2] = \mathrm{E}u^2 + 2(\mu-a)\underbrace{\mathrm{E}u}_{0} + (\mu-a)^2 = \mathrm{E}u^2 + (\mu-a)^2$.

$\mathrm{E}u^2$ does not depend on a, and $(\mu-a)^2 \geq 0$ is minimized at $a = \mu$. ■

Card 2 — L8 Q2 · CEF is the best predictor

$m(x) = \mathrm{E}(y|x)$, $y = m(x) + u$. (a) Show $\mathrm{E}(u|x) = 0$. (b) Show $\mathrm{E}(h(x)u) = 0$. (c) Show $g(x) = m(x)$ minimizes $\mathrm{E}[(y-g(x))^2]$.

Hook: (a) definition; (b) LIE; (c) same trick as Card 1 with $h = m - g$.

(a) $\mathrm{E}(u|x) = \mathrm{E}(y|x) - m(x) = m(x) - m(x) = 0$.
(b) $\mathrm{E}(h(x)u) = \mathrm{E}_x\big[\mathrm{E}(h(x)u|x)\big] = \mathrm{E}_x\big[h(x)\mathrm{E}(u|x)\big] = 0$ (LIE, then (a)).
(c) $\mathrm{E}[(y-g)^2] = \mathrm{E}[(u+m-g)^2] = \mathrm{E}u^2 + 2\mathrm{E}[u\,(m-g)] + \mathrm{E}(m-g)^2$. Set $h = m - g$: the middle term is 0 by (b). So $\mathrm{E}u^2 + \mathrm{E}(m-g)^2$, minimized at $g(x) = m(x)$. ■

Card 3 — L8 Q3 · residual facts

FOCs $-2\sum_i x_{ik}(y_i - x_i'b) = 0\ \forall k$, $e_i = y_i - x_i'b$, $x_{i1} = 1$. Show (a) $\sum e_i = 0$, (b) $\bar{y} = \bar{x}'b$, (c) $\widehat{\mathrm{Cov}}(x_k, e) = 0$.

Hook: FOC $= \sum x_{ik}e_i = 0$; plug in $k = 1$; average; expand the covariance.

FOC $\Rightarrow \sum_i x_{ik}e_i = 0$ for every k. (a) $k = 1$, $x_{i1} = 1$: $\sum_i e_i = 0$. (b) $0 = \bar{e} = \frac{1}{n}\sum(y_i - x_i'b) = \bar{y} - \bar{x}'b \Rightarrow \bar{y} = \bar{x}'b$.
(c) $\frac{1}{n}\sum(x_{ik} - \bar{x}_k)(e_i - \bar{e}) = \frac{1}{n}\sum x_{ik}e_i - \bar{x}_k\bar{e} = 0 - 0 = 0$ (FOC and (a)). ■

Card 4 — L8 Q4 · bivariate OLS, slope = Cov/Var

$X = [\iota \;\; e]$ (education), $Y = w$. (a) Show $b = \begin{bmatrix} 1 & \bar{e} \\ \bar{e} & \overline{ee} \end{bmatrix}^{-1} \begin{bmatrix} \bar{w} \\ \overline{ew} \end{bmatrix}$, showing the multiplication. (b) Show $b_2 = \widehat{\mathrm{Cov}}(e,w)/\widehat{\mathrm{Var}}(e)$.

Hook: multiply out, factor n, n's cancel; 2×2 inverse, read row 2.

(a) $X'X = \begin{bmatrix} 1 & \cdots & 1 \\ e_1 & \cdots & e_n \end{bmatrix} \begin{bmatrix} 1 & e_1 \\ \vdots & \vdots \\ 1 & e_n \end{bmatrix} = \begin{bmatrix} n & \sum e_i \\ \sum e_i & \sum e_i^2 \end{bmatrix} = n \begin{bmatrix} 1 & \bar{e} \\ \bar{e} & \overline{ee} \end{bmatrix}$, $X'Y = \begin{bmatrix} \sum w_i \\ \sum e_i w_i \end{bmatrix} = n \begin{bmatrix} \bar{w} \\ \overline{ew} \end{bmatrix}$.

$b = \frac{1}{n} \begin{bmatrix} 1 & \bar{e} \\ \bar{e} & \overline{ee} \end{bmatrix}^{-1} n \begin{bmatrix} \bar{w} \\ \overline{ew} \end{bmatrix}$, and the n's cancel.

(b) $b = \dfrac{1}{\overline{ee} - \bar{e}^2} \begin{bmatrix} \overline{ee} & -\bar{e} \\ -\bar{e} & 1 \end{bmatrix} \begin{bmatrix} \bar{w} \\ \overline{ew} \end{bmatrix}$, so row 2: $b_2 = \dfrac{\overline{ew} - \bar{e}\bar{w}}{\overline{ee} - \bar{e}^2} = \dfrac{\widehat{\mathrm{Cov}}(e,w)}{\widehat{\mathrm{Var}}(e)}$. ■

Card 5 — L9 Q1 · OLS is unbiased

$Y = X\beta + \boldsymbol{\varepsilon}$, $\mathrm{E}(\boldsymbol{\varepsilon}|X) = 0$, $b = (X'X)^{-1}X'Y$. Show (a) $\mathrm{E}(b|X) = \beta$, (b) $\mathrm{E}(b) = \beta$.

Hook: substitute Y; pull out the functions of X; A3; LIE.

(a) $b = (X'X)^{-1}X'(X\beta + \boldsymbol{\varepsilon}) = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon}$. $\mathrm{E}(b|X) = \beta + (X'X)^{-1}X'\,\mathrm{E}(\boldsymbol{\varepsilon}|X) = \beta + 0 = \beta$ (functions of X are constants given X).
(b) LIE: $\mathrm{E}(b) = \mathrm{E}_X[\mathrm{E}(b|X)] = \mathrm{E}_X[\beta] = \beta$. ■

Card 6 — L9 Q2 · variance sandwich $\to \sigma^2(X'X)^{-1}$

(a) Show $b = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon}$. (b) Show $\mathrm{Var}(b|X) = (X'X)^{-1}X'\mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'|X]X(X'X)^{-1}$. (c) Why is $\mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'|X] = \mathrm{Var}(\boldsymbol{\varepsilon}|X)$? (d) Homoskedasticity + no autocorrelation $\Rightarrow \sigma^2 I \Rightarrow \mathrm{Var}(b|X) = \sigma^2(X'X)^{-1}$.

Hook: $b - \beta = A\boldsymbol{\varepsilon}$; sandwich $A(\cdot)A'$; mean zero; diagonal σ^2, off-diagonal 0.

(a) $b = (X'X)^{-1}X'(X\beta + \boldsymbol{\varepsilon}) = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon}$.
(b) $b - \mathrm{E}(b|X) = (X'X)^{-1}X'\boldsymbol{\varepsilon}$, so $\mathrm{Var}(b|X) = \mathrm{E}\big[(X'X)^{-1}X'\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'X(X'X)^{-1}\,|X\big] = (X'X)^{-1}X'\mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'|X)X(X'X)^{-1}$.
(c) $\mathrm{Var}(\boldsymbol{\varepsilon}|X) = \mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'|X) - \mathrm{E}(\boldsymbol{\varepsilon}|X)\mathrm{E}(\boldsymbol{\varepsilon}|X)'$, and the second term is 0 since $\mathrm{E}(\boldsymbol{\varepsilon}|X) = 0$.
(d) Homoskedastic: every diagonal entry $\mathrm{Var}(\varepsilon_i|X) = \sigma^2$. No autocorrelation: off-diagonal $\mathrm{Cov}(\varepsilon_i, \varepsilon_j|X) = 0$. So $\sigma^2 I_n$, and
$\mathrm{Var}(b|X) = (X'X)^{-1}X'(\sigma^2 I)X(X'X)^{-1} = \sigma^2(X'X)^{-1}\underbrace{X'X(X'X)^{-1}}_{I} = \sigma^2(X'X)^{-1}$. ■

Card 7 — **L9 Q3 · LLN ⇒ consistent, CLT ⇒ normal**

$b = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon}$, $w_i = x_i\varepsilon_i$. (a) Show $b = \beta + (X'X/n)^{-1}\bar{w}$. (b) What does the LLN imply? (c) What does the CLT imply?

Hook: $X'\boldsymbol{\varepsilon} = n\bar{w}$; LLN sends averages to means; CLT makes averages normal.

(a) $X'\boldsymbol{\varepsilon} = \sum_i x_i\varepsilon_i = n\bar{w}$, so $b = \beta + (X'X)^{-1}n\bar{w} = \beta + \left(\frac{X'X}{n}\right)^{-1}\bar{w}$.

(b) LLN: $\bar{w} \to \mathrm{E}(w_i) = 0$ and $\frac{X'X}{n} \to \mathrm{E}(x_i x_i')$ (invertible), so $\operatorname{plim} b = \beta$. OLS is **consistent**.

(c) CLT: $\bar{w}$ is approximately normal in large samples, and b is linear in $\bar{w}$, so $b \overset{a}{\sim} N(\beta, \sigma^2(X'X)^{-1})$ **even without normal errors**. This justifies t-tests and CIs.

Card 8 — **L10 Q1 · $P_X H = 0$, regressing e on X gives $a = 0$**

$H = (e_1, \ldots, e_n)'$. (a) Define P_X. (b) Show $P_X H = 0$. (c) Say (b) in words. (d) Regress e on X; show $a = 0$.

Hook: $P_X Y = Xb$, $P_X X = X$, so $Xb - Xb$; then $b - b$.

(a) $P_X = X(X'X)^{-1}X'$ $(n \times n)$.

(b) $P_X H = X(X'X)^{-1}X'(Y - Xb) = X\underbrace{(X'X)^{-1}X'Y}_{b} - X\underbrace{(X'X)^{-1}X'X}_{I}b = Xb - Xb = 0.$

(c) The residuals are orthogonal to every column of X: no linear combination of the regressors can predict them.

(d) $a = (X'X)^{-1}X'H = (X'X)^{-1}X'Y - (X'X)^{-1}X'Xb = b - b = 0$. ■

Card 9 — **L10 Q2 · projecting on ones = averaging**

$Y = \tau Z + \kappa X + \boldsymbol{\varepsilon}$ with Z a column of ones. Describe completely (dimensions, every element): (a) $P_Z = Z(Z'Z)^{-1}Z'$, (b) $P_Z X$, (c) X^* in $M_Z Y = \kappa M_Z X + M_Z\boldsymbol{\varepsilon}$.

Hook: $Z'Z = n$, so all entries are $1/n$; average; demean.

(a) $Z'Z = n$, so $P_Z = \frac{1}{n}ZZ'$: an $n \times n$ matrix with **every element** $1/n$.

(b) $P_Z X$ is $n \times 1$; each row is $\sum_j \frac{1}{n}x_j = \bar{x}$, so **every element is** $\bar{x}$ (the average price).

(c) $X^* = M_Z X = X - P_Z X$ is $n \times 1$ with element i equal to $x_i - \bar{x}$: demeaned price.

Card 10 — L11 Q1 · standardized estimate: $N(0,1)$, then t_{n-K}

$b|X \sim N(\beta, \sigma^2(X'X)^{-1})$. Explain how the standardized estimate is built and give its distribution.

Hook: take the kth element, subtract the mean, divide by the SD; swap σ^2 for s^2 and you get t.

$b_k|X \sim N\big(\beta_k, \sigma^2[(X'X)^{-1}]_{kk}\big)$, so $z_k = \dfrac{b_k - \beta_k}{\sqrt{\sigma^2[(X'X)^{-1}]_{kk}}} \sim N(0,1)$.

σ^2 is unknown, so use $s^2 = \dfrac{e'e}{n-K}$: $t_k = \dfrac{b_k - \beta_k}{\sqrt{s^2[(X'X)^{-1}]_{kk}}} = \dfrac{b_k - \beta_k}{\widehat{SE}(b_k)} \sim t_{n-K}$.

Use: CI $= b_k \pm t_{crit}\,\widehat{SE}$; to test $H_0 : \beta_k = q$, use $t = (b_k - q)/\widehat{SE}$.

Card 11 — L11 Q2 · $e'e = \varepsilon' M \varepsilon$ and s^2 unbiased (no shortcuts)

$M_X = I - X(X'X)^{-1}X'$, $e = M_X\varepsilon$. (a) Show $e'e = \varepsilon' M_X \varepsilon$. (b) Given $\mathrm{E}(e'e|X) = \mathrm{tr}(M_X \mathrm{E}(\varepsilon\varepsilon'|X))$, show $\mathrm{E}\big(\frac{e'e}{n-K}\big|X\big) = \sigma^2$.

Hook: M symmetric + idempotent; $\sigma^2 I$ comes out; $\mathrm{tr}M = n - K$ by cycling.

(a) $e'e = \varepsilon' M' M \varepsilon$. Symmetric: $M' = I - X((X'X)^{-1})'X' = M$ (since $X'X$ is symmetric). Idempotent: $MM = I - 2X(X'X)^{-1}X' + X\underbrace{(X'X)^{-1}X'X}_{I}(X'X)^{-1}X' = I - 2P + P = I - P = M$. So $e'e = \varepsilon' M \varepsilon$.

(b) A4: $\mathrm{E}(\varepsilon\varepsilon'|X) = \sigma^2 I_n$, so $\mathrm{E}(e'e|X) = \mathrm{tr}(M\sigma^2 I_n) = \sigma^2 \mathrm{tr}(M)$.
$\mathrm{tr}(M) = \mathrm{tr}(I_n) - \mathrm{tr}(X(X'X)^{-1}X') = n - \mathrm{tr}(\underbrace{(X'X)^{-1}X'X}_{I_K}) = n - K$ (cycle the trace).

So $\mathrm{E}(e'e|X) = \sigma^2(n-K)$, and dividing by $n-K$ gives $\mathrm{E}\big(\frac{e'e}{n-K}\big|X\big) = \sigma^2$. ■

Card 12 — L11 Q3 · bivariate $\mathrm{Var}(b_2) = \sigma^2 / \sum(x_i - \bar{x})^2$

$y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$, $\mathrm{Var}(b|X) = \sigma^2(X'X)^{-1}$. (a) Derive $\mathrm{Var}(b_2|X)$. (b) Name the three factors.

Hook: write out $X'X$; take the $(2,2)$ entry of the inverse; the determinant equals $n\,SST_x$.

(a) $X'X = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}$, so $(X'X)^{-1} = \dfrac{1}{n\sum x_i^2 - (\sum x_i)^2}\begin{bmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{bmatrix}$.

$(2,2)$ entry: $\mathrm{Var}(b_2|X) = \dfrac{\sigma^2 n}{n\sum x_i^2 - n^2\bar{x}^2} = \dfrac{\sigma^2}{\sum x_i^2 - n\bar{x}^2} = \dfrac{\sigma^2}{\sum(x_i - \bar{x})^2} = \dfrac{\sigma^2}{(n-1)s_x^2}$.

(b) Larger σ^2 (noise) ↑ variance; larger n ↓ variance; more spread in x (s_x^2) ↓ variance.

Card 13 — **L11 Q4 · multiple regression** $\mathrm{Var}(b_3)$

$y_i = \beta_1 + \beta_2 x_{i2} + \cdots + \beta_K x_{iK} + \varepsilon_i$. (a) Write $\mathrm{Var}(b_3|X)$. (b) Explain each of the 3 components and when it improves precision.

Hook: "sigma squared over (one minus R-three-squared) times SST-three."

(a) $\mathrm{Var}(b_3|X) = \dfrac{\sigma^2}{(1-R_3^2)\,SST_3}$, where $SST_3 = \sum_i (x_{i3} - \bar{x}_3)^2$ and R_3^2 is the R^2 from regressing x_3 on all the other regressors.
(b) σ^2: variance of the unobservables; smaller (e.g. add controls that explain y) $\Rightarrow$ more precise.
SST_3: total variation in x_3; more variation or a larger n $\Rightarrow$ more precise.
$1 - R_3^2$: share of x_3's variation *not* explained by the other x's; less correlation with the others $\Rightarrow$ more precise.

Card 14 — **L11 Q5 · multicollinearity and the bias–variance trade-off**

(a) What is multicollinearity and how does it affect precision? (b) Why can dropping a highly correlated regressor improve precision but cause bias?

Hook: $R_k^2 \uparrow \Rightarrow$ Var $\uparrow$; dropping it lowers the variance but puts the variable in ε, so A3 fails and you get OVB.

(a) Multicollinearity means x_k is highly correlated with the other regressors (high R_k^2). Then $1 - R_k^2 \to 0$, so $\mathrm{Var}(b_k) = \frac{\sigma^2}{(1-R_k^2)SST_k}$ rises: large SEs and imprecise (but still unbiased) estimates.
(b) Dropping the correlated x_j lowers R_k^2, so $\mathrm{Var}(b_k)$ falls. But x_j moves into the error, and since it is correlated with x_k, $\mathrm{E}(\varepsilon|X) \neq 0$ (A3 fails). The result is **omitted variable bias**, $\mathrm{E}(b_1) - \beta_1 = \Gamma\beta_2 \neq 0$. You trade lower variance for bias.

Card 15 — **L12 Q1 · log-level:** $\log(fare) = 4.666 + 0.043\,dist$ **(*dist* in 100s of miles)**

(a) Write *fare* in terms of *dist*. (b) Find $\partial fare/\partial dist$. (c) Show 1 mile $\Rightarrow \approx 0.043\%$ higher fare.

Hook: exponentiate; chain rule gives $\beta \cdot fare$; 1 mile is $\partial dist = 0.01$.

(a) $fare = e^{4.666 + 0.043\,dist}$. (b) $\dfrac{\partial fare}{\partial dist} = 0.043\,e^{4.666+0.043\,dist} = 0.043\,fare$.
(c) $100\dfrac{\partial fare}{fare} = 4.3\,\partial dist$. One mile $= \partial dist = 0.01$, so $100\frac{\partial fare}{fare} = 4.3(0.01) = 0.043\%$. ■

Card 16 — **L12 Q2** · **level-log:** $fare = 41 + 66\log(dist)$

(a) Find $\partial fare/\partial dist$. (b) Show a 1% increase in $dist$ $\Rightarrow$ +\$0.66. (c) Show this does not depend on the units of $dist$ (e.g. miles).

Hook: β/x; multiply and divide by 100; the $\log 100$ goes into the intercept.

(a) $\dfrac{\partial fare}{\partial dist} = \dfrac{66}{dist}$. (b) $\partial fare = 66\dfrac{\partial dist}{dist} = \dfrac{66}{100}\left(100\dfrac{\partial dist}{dist}\right) = 0.66 \times (\%\Delta dist)$, so $+1\% \Rightarrow$ +\$0.66.
(c) In miles, $m = 100\,dist$: $fare = 41 + 66\log(m/100) = (41 - 66\log 100) + 66\log m$. Only the intercept changes; the slope is still 66, so still \$0.66. ■

Card 17 — **L12 Q3** · **log-log:** $\log(fare) = 4.25 + 0.40\log(dist)$

(a) Write $fare$ in terms of $dist$. (b) Find $\partial fare/\partial dist$. (c) Show a 10% increase in $dist$ $\Rightarrow\approx$4% higher fare.

Hook: log-log means elasticity: $\%\Delta y = \beta \cdot \%\Delta x$.

(a) $fare = e^{4.25}\,dist^{0.40}$. (b) $\dfrac{\partial fare}{\partial dist} = 0.40\,e^{4.25}dist^{-0.60} = 0.40\,\dfrac{fare}{dist}$.
(c) Rearrange: $100\dfrac{\partial fare}{fare} = 0.40\left(100\dfrac{\partial dist}{dist}\right) = 0.40 \times 10 = 4\%$. ■

Last-minute hooks (cover the answers and recite these)

1 mean / 2 CEF	add and subtract the truth, the cross term dies (LIE for the CEF)
3 residuals	FOC $\sum x_{ik}e_i = 0 \to k$=1 $\to$ average $\to$ Cov = FOC $-\bar{x}\bar{e}$
4 bivariate b	multiply out, factor n, 2×2 inverse, row 2
5 unbiased	substitute Y, A3, LIE
6 variance	$A\boldsymbol{\varepsilon} \to A\mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}')A' \to \sigma^2 I \to$ cancel
7 LLN/CLT	$X'\boldsymbol{\varepsilon} = n\bar{w}$; consistent; approximately normal without A5
8 $P_X H$	$Xb - Xb$; $b - b$
9 P_Z	all entries $1/n \to \bar{x} \to x_i - \bar{x}$
10 t-stat	$N(0,1)$ with σ; t_{n-K} with s
11 s^2	$M' = M$, $MM = M$, $\mathrm{tr}M = n - K$
12/13 $\mathrm{Var}(b_k)$	σ^2/SST_x; $\sigma^2/[(1 - R_k^2)SST_k]$; 3 factors
14 collinearity	$R_k^2 \uparrow\Rightarrow$ Var $\uparrow$; drop $\Rightarrow$ OVB
15–17 logs	log-level 100β%; level-log $\beta/100$; log-log elasticity β